Appendix 1. Search Strategy

Table 1. Medline Search 1946 to October 23, 2020

#	Searches	Results
1	exp comorbidity/ or multiple chronic conditions/	111191
2	(multimorbid* or multi-morbid* or comorbid* or co-morbid*).ab,ti,kw.	180089
3	((multiple or coexist* or co-exist* or concurrent* or simultaneous*) adj2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)).ab,ti,kw.	52139
4	1 or 2 or 3	292316
5	primary health care/ or "continuity of patient care"/ or exp general practice/ or ambulatory care/ or physicians, family/ or physicians, primary care/ or community health services/ or general practitioners/	249484
6	((ambulatory or community or general or family or primary) adj2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*)).ab,ti,kw.	391013
7	5 or 6	508037
8	epidemiologic measurements/ or risk assessment/ or "Outcome Assessment (Health Care)"/ or patient reported outcome measures/ or health status indicators/ or "severity of illness index"/ or sickness impact profile/ or diagnosis-related groups/ or case mix/	594279
9	((health status or risk or outcome* or sickness impact) adj2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*)).ab,ti,kw.	398319
10	((severity or burden) adj2 (illness* or diseas* or disorder* or condition* or diagnos*)).ab,ti,kw.	85595
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix).ab,ti,kw.	21547
12	8 or 9 or 10 or 11	939031
13	4 and 7 and 12	4862
14	limit 13 to (english language and yr="2010 - Current")	2827

Table 2. Embase Classic+Embase 1947 to October 23, 2020

#	Searches	Results
1	comorbidity/ or multiple chronic conditions/	265602
2	(multimorbid* or multi-morbid* or comorbid* or co-morbid*).ab,ti,kw.	329397
3	((multiple or coexist* or co-exist* or concurrent* or simultaneous*) adj2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)).ab,ti,kw.	81506
4	1 or 2 or 3	493137
5	primary health care/ or patient care/ or primary medical care/ or general practice/ or general practitioner/ or ambulatory care/ or family medicine/ or community care/ or community health services/	664384
6	((ambulatory or community or general or family or primary) adj2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*)).ab,ti,kw.	539525
7	5 or 6	943743
8	risk assessment/ or outcome assessment/ or patient reported outcome/ or health status indicator/ or disease activity score/ or global disease burden/ or organ dysfunction score/ or "severity of illness index"/ or sickness impact profile/ or general health status assessment/ or disease severity/ or diagnosis related group/ or charlson comorbidity index/ or comorbidity assessment/ or elixhauser comorbidity index/ or case mix/	1590115
9	((health status or risk or outcome* or sickness impact) adj2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*)).ab,ti,kw.	547083
10	((severity or burden) adj2 (illness* or diseas* or disorder* or condition* or diagnos*)).ab,ti,kw.	138554
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix).ab,ti,kw.	42616
12	8 or 9 or 10 or 11	2022565
13	4 and 7 and 12	12245
14	limit 13 to (english language and yr="2010 - Current")	8896

Table 3. CINAHL – since inception to October 23, 2020

#	Searches	Results
1	(MH “comorbidity”)	38970
2	(multimorbid* or multi-morbid* or comorbid* or co-morbid*)	66442
3	(multiple or coexist* or co-exist* or concurrent* or simultaneous*) N2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)	12822
4	1 or 2 or 3	77266
5	((MH “primary health care”) or (MH “family practice”) or (MH “ambulatory care") or (MH “ambulatory care facilities") or (MM “community health services”) or (MM “community health centers”) or (MH “physicians, family”) or (MH “continuity of patient care"))	100224
6	((ambulatory or community or general or family or primary) N2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*))	332346
7	5 or 6	340014
8	((MH “risk assessment”) or (MH "Outcome Assessment") or (MH “patient-reported outcomes) or (MH “health status indicators”) or (MH "severity of illness indices") or (MH “sickness impact profile”) or (MH “diagnosis-related groups”) or (MH “case mix”))	102350
9	((health status or risk or outcome* or sickness impact) N2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*))	340802
10	(severity or burden) N2 (illness* or diseas* or disorder* or condition* or diagnos*)	68742
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix)	6729
12	8 or 9 or 10 or 11	396387
13	4 and 7 and 12	4729
14	limit 13 to (english language and yr="2010 - Current")	3536

Appendix 2. Coding Description for the Modified Newcastle-Ottawa Scale for Cohort Studies

Category	Description
Selection	
1) Representativeness of the sample	This item assesses the representativeness of sample in the community, not from some general population. a) Truly representative* (e.g., everyone from the database) b) Somewhat representative* (with at least 2 criteria but selection method was convincing due to random sampling) c) Selected group (e.g., only certain socio-economic groups or areas) d) No description of sampling strategy
2) Ascertainment of multimorbidity	This item assesses the method by which multimorbidity was confirmed. a) Secure record* (e.g., GP questionnaire) b) Structured interview* (e.g., interviewer-administered questionnaire) c) Written self-report (e.g., mailed survey, if items are unable to be confirmed by objective measure) d) No description / Other
3) Demonstration that outcome of interest was not present at start of study	A statement of no history of disease earns a star. In the case of mortality studies, outcome of interest is still the presence of a disease, rather than death. a) Yes* b) No
Comparability	
1) Study controls for age and sex	Covariates must be adjusted for in the analysis. Statements of no differences between groups or that differences were not statistically significant are not sufficient for establishing comparability. Note: If the relative risk for the exposure of interest is adjusted for the covariates listed, then the groups will be considered to be comparable on each variable used. a) Yes* b) No
2) Study controls for other factors	a) Yes* b) No
Outcome	
1) Statistical test	Statistical test(s) must be clearly described and appropriate before assessing the other items under the Outcome category. a) Clearly described and appropriate b) Not described, incomplete or inappropriate
2) Assessment of outcome	This item assesses the method by which the outcome of interest was confirmed. For some outcomes, reference to the medical record is sufficient to satisfy the requirement for confirmation. This may not be adequate for other outcomes where reference to specific tests or measures would be required. a) Independent or blind assessment* (e.g., interviewer-administered questionnaire) b) Record linkage* (e.g., identified through ICD codes on database records) c) Self-report (i.e., no reference to original medical records) d) No description / Other

3)	Was follow-up long enough for outcomes to occur?	An acceptable length of time was at least 1 year of follow-up. a) Yes***** b) No
4)	Adequacy of follow-up of cohorts	This item assesses the follow-up of the sample to ensure that losses are not related to the outcome. a) Complete follow-up - all subject accounted for***** b) Subjects lost to follow-up unlikely to introduce bias***** (Number lost ≤20% or description of those lost suggested no different from those followed.) c) Follow-up rate less than 80% and no description of those lost d) No statement

Thresholds for converting the modified Newcastle-Ottawa scales to Good, Fair, and Poor Quality:

- Good Quality - 2-3 stars in Selection category (Representativeness of Sample item must be fulfilled) AND 1-2 stars in Comparability category AND 2-3 stars in Outcome category.
- Fair Quality - 1 star in Selection category AND 1-2 stars in Comparability category AND 2-3 stars in Outcome category.
- Poor Quality - 0 star in Selection category OR 0 star in Comparability category OR 0-2 stars in Outcome category.

Wells GA, Shea B, O'Connell D, Peterson J, Welch V, Losos M, Tugwell P. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomized studies in meta-analysis. Available from: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp

Modifications from original NOS include:

- 'Statistical test' was added to the Outcome category
- 'Representativeness of sample' item under Selection category and 'Statistical test' item under Outcome category must both be fulfilled for study to be considered Good Quality.
- 'Selection of the non-exposed cohort' item was removed from Selection category as most studies did not describe a non-exposed cohort, and this review sought to compare the different levels of multimorbidity within the group.
- 'Representativeness of the exposed cohort' item under Selection category was renamed 'Representativeness of the sample' since the 'non-exposed cohort' was removed above.
- 'Ascertainment of exposure' under the Selection category was renamed as 'Ascertainment of multimorbidity' to specify that multimorbidity is the exposure in this review.
- 'Study controls for age and sex' and 'Study controls for other factors' were revised to be items under Comparability category
- The thresholds for converting the scales to quality were amended accordingly due to the above modifications.

Consensus:

- The study team decided that a period of one-year was a reasonable period of follow-up under 'Was follow-up long enough for outcomes to occur?' item under Outcome category.

Appendix 3. Coding Description for the Modified Newcastle-Ottawa Scale for Cross-sectional Studies

Category	Description
Selection	
1) Representativeness of the sample	This item assesses the representativeness of sample in the specified population, not from some general population. a) Truly representative* (e.g., everyone from the database, random sampling) b) Somewhat representative* (with at least 2 criteria but selection method was convincing due to random sampling) c) Selected group (e.g., only certain socio-economic groups or areas) d) No description of sampling strategy
2) Ascertainment of multimorbidity	This item assesses the method by which multimorbidity was confirmed. a) Secure record* (e.g., Clinical records, GP questionnaire) b) Structured interview* (e.g., interviewer-administered questionnaire) c) Written self-report (e.g., mailed survey, if items are unable to be confirmed by objective measure) d) No description / Other
3) Sample Size	If there is no description, a reported sample size of 800 and above is satisfactory. a) Justified and satisfactory* b) Not justified
4) Non-respondents	Acceptable response rates for surveys through various methods† • In-person: 57% • Mail: 50% • Average: 33% • Email: 30% • Internet: 29% • Telephone: 18% • In-app: 13% a) Comparability between respondents and non-respondents characteristics is established, and the response rate is satisfactory.* b) The response rate is unsatisfactory, or the comparability between respondents and non-respondents is unsatisfactory. c) No description of the response rate or the characteristics of the responders and the non-responders
Comparability	
1) Study controls for age and sex	Confounders must be adjusted for in the analysis. Statements of no differences between groups or that differences were not statistically significant are not sufficient for establishing comparability. Note: If the relative risk for the exposure of interest is adjusted for the confounders listed, then the groups will be considered to be comparable on each variable used in the adjustment. a) Yes* b) No
2) Study controls for other factors	a) Yes* b) No

Outcome	
1) Statistical test	Statistical test(s) must be clearly described and appropriate before assessing the other items under the Outcome category. a) Clearly described and appropriate b) Not described, incomplete or inappropriate
2) Assessment of outcome	This item assesses the method by which the outcome of interest was confirmed. For some outcomes, reference to the medical record is sufficient to satisfy the requirement for confirmation. This may not be adequate for other outcomes where reference to specific tests or measures would be required. a) Independent or blind assessment***** b) Record linkage***** (e.g., identified through ICD codes on database records) c) Self-report (i.e., no reference to original medical records) d) No description / Other

Thresholds for converting the modified Newcastle-Ottawa scales to Good, Fair, and Poor Quality:

- Good Quality – 3-4 stars in Selection category (Representativeness of Sample item must be fulfilled) AND 1-2 stars in Comparability category AND 1 star in Outcome category.
- Fair Quality - 2 stars in Selection category AND 1-2 stars in Comparability category AND 1 star in Outcome category.
- Poor Quality – 0-1 star in Selection category OR 0 star in Comparability category OR 0 star in Outcome category.

†Lindermann N. What's the average survey response rate? [2018 benchmark]. SurveyAnyplace. Available from: https://surveyanyplace.com/average-survey-response-rate/.

Alshabanat A, Zafari Z, Albanyan O, Dairi M, FitzGerald JM. Asthma and COPD Overlap Syndrome (ACOS): a systematic review and meta-analysis. PLoS ONE. 2015;10(9):e0136065. Doi:10.1371/journal.pone.0136065.

Modifications from original NOS include:

- 'Ascertainment of exposure' item under Selection category was renamed as 'Ascertainment of multimorbidity' to specify that multimorbidity is the exposure in this review.
- The ratings for the 'Ascertainment of exposure' item under Selection category were revised to 'secure record, structured interview, written self-report, no description/other' to align with the Modified NOS for cohort studies in this review.
- 'Study controls for age and sex' and 'Study controls for other factors' items were revised to be items under Comparability category.
- 'Was follow-up long enough for outcomes to occur' item under Outcome category was renamed as 'Statistical test'. This item must be fulfilled for the study to be considered as Good Quality.
- The thresholds for converting the scales to quality were amended accordingly due to the above modifications.

Consensus:

- The study team decided on a list of acceptable response rates for various survey methods for the 'Non-respondents' item under Selection category.
- The study team decided that a sample size of 800 was reasonable under 'Sample size' item under Selection category.

Appendix 4. Summary of risk of bias (RoB) appraisal of included cohort articles using the modified Newcastle-Ottawa Quality Assessment Scale (44 articles)

	Barile et al. (2013)[19]	Biehl et al. (2016)[22]	Boeckxstaens et al. (2015b)[24]	Brilleman et al. (2014)[25]	Brilleman and Salisbury(2013)[26]	Carey et al. (2013)[28]	Chapman et al. (2015)[29]	Charlson et al. (2014)[30]	Crane et al. (2010)[35]	Crooks et al. (2016)[36]	Crooks et al. (2015)[37]	Formiga et al. (2013)[39]	Formiga et al. (2011a)[40]	Formiga et al. (2016)[42]	Fraccaro et al. (2016)[43]	Haas et al. (2013)[48]	Hwang et al. (2015)[51]	Jennings et al. (2015)[53]	Jia et al. (2018)[54]	Jia and Lubetkin (2017)[55]	Lemke et al. (2012)[62]	Md Yusof et al. (2010)[68]	Monterde et al. (2020)[70]	Pavne et al. (2013)[77]	Pavne et al. (2014)[78]	Pavne et al. (2020)[79]	Quail et al. (2011)[81]	Reves et al. (2014)[85]	Salisbury et al. (2011)[87]	Saver et al. (2014)[88]	Stanley and Sarfati (2017)[91]	St John et al. (2014)[92]	St John et al. (2019)[93]	Streit et al. (2014)[94]	Takahashi et al. (2011)[96]	Takahashi et al. (2016)[97]	Tvack et al. (2016)[98]	Wallace et al. (2016a)[103]	Wallace et al. (2016b)[104]	Wei et al. (2019a)[106]	Wei et al. (2019b)[107]	Wei et al. (2020a)[108]	Wei et al. (2020b)[109]	Wei and Mukamal (2018)[110]
Selection Criteria																																												
Representativeness of the sample	*	*	-	*	*	*	*	-	*	*	*	*	*	*	*	*	*	-	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	-	*	*	*
Ascertainment of Multimorbidity	-	*	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	-	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	-	*	*	-
Demonstration that outcome of interest was not present at the start of the study	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Comparability Criteria																																												
Study controls for age and sex	*	-	*	*	*	*	*	*	-	*	*	-	*	-	*	*	*	*	-	*	*	-	*	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	-	*	-	-
Study controls for others	*	*	-	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	-	-	*	*	*	*	*	-	*	*	*	*	-	*	*	*	*	*	*	*	*	*	-	*
Outcome Criteria																																												
Statistical test†	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Assessment of outcome	-	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Follow-up was long enough for outcomes to occur	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Adequacy of follow-up of cohorts	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	-	*	*	*	*	-	*	*	*	-	-	*	*
Overall RoB Judgement[a]	G	G	F	G	G	G	G	G	G	G	G	G	G	G	G	G	G	G	F	P	P	G	F	G	G	G	G	G	G	G	G	G	G	G	G	G	F	G	G	G	F	G	P	G

Note. G = Good; F = Fair; P = Poor.

[†]Statistical test must be clearly described and appropriate before assessing the other items under the Outcome category. [a] Good rating is given when there are 2 to 3 stars in Selection category (Representativeness of Sample must be fulfilled) AND 1 to 2 stars in Comparability category AND 2 to 3 stars in Outcome category; Fair rating is given when there are 1 star in Selection category AND 1 to 2 stars in Comparability category AND 2 to 3 stars in Outcome category; Poor rating is given when there are 0 star in Selection category OR 0 star in Comparability category OR 0 to 2 stars in Outcome category.

"*" Study satisfies the criteria; "-" Study did not satisfy the criteria; "✓" Statistical test is clearly described and appropriate; "X" Statistical test is not described, incomplete or inappropriate.

Appendix 5. Summary of risk of bias (RoB) appraisal of included cross-sectional articles using the modified Newcastle-Ottawa Quality Assessment Scale (52 articles)

	Agborsangaya et al. (2013)[17]	Bahler et al. (2015)[18]	Barile et al. (2012)[20]	Barnett et al. (2012)[21]	Boeckxstaens et al. (2015a)[23]	Caballer-Tarazona et al. (2019)[27]	Chen et al. (2011)[31]	Chen et al. (2018)[32]	Chu (2018)[33]	Clynes et al. (2020)[34]	DiNapoli et al. (2017)[38]	Formiga et al. (2011b)[41]	Galenkamp et al. (2011)[44]	Garin et al. (2014)[45]	Givnn et al. (2011)[46]	Gunn et al. (2012)[47]	Hanmer et al. (2010)[49]	Hu et al. (2017)[50]	Isaacs et al. (2014)[52]	Jindai et al. (2016)[56]	Kim et al. (2012)[57]	Koiima et al. (2011)[58]	Kristensen et al. (2014)[59]	Lani et al. (2015)[60]	Lawson et al. (2013)[61]	Li et al. (2016)[63]	Lonrinzi (2016)[64]	Macinko et al. (2019)[65]	Marengoni et al. (2011)[66]	McDaid et al. (2013)[67]	Milla-Perseguer (2019)[69]	Murgah et al. (2012)[71]	Mujica-Mota et al. (2015)[72]	Naessens et al. (2011)[73]	Østergaard and Foldager (2011)[74]	Palladino et al. (2019)[75]	Pati et al. (2019)[76]	Peters et al. (2018)[80]	Ranstad et al. (2014)[82]	Reinke et al. (2019)[83]	Renne and Gobbens (2018)[84]	Ryu et al. (2015)[86]	Shadmi et al. (2011)[89]	Sibley et al. (2014)[90]	Sullivan et al. (2012)[95]	Ubalde-Lopez et al. (2016)[99]	van den Bussche et al. (2014)[100]	van Oostrom et al. (2014)[101]	Vos et al. (2013)[102]	Wei et al. (2018)[105]	Wikman et al. (2011)[111]	Wister et al. (2015)[112]
Selection																																																					
Representativeness of the sample	*	*	*	*	-	*	*	-	*	-	*	*	*	*	*	*	*	-	*	-	*	-	*	*	*	*	*	*	*	*	*	*	-	-	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*
Ascertainment of Multimorbidity	*	*	-	*	*	*	*	-	*	*	*	*	*	*	*	*	-	-	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*	-	*	*	-	*	*	*	-	-	*	*	*	*	*	*	*	*	*
Sample size	*	*	*	*	-	*	*	*	*	-	*	-	*	*	*	*	*	*	-	*	*	*	-	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*
Non-respondents	-	*	*	*	-	*	*	*	*	-	*	*	-	-	*	-	-	*	*	*	*	-	*	*	*	-	*	-	-	-	-	*	-	*	-	*	*	-	*	*	-	-	*	*	*	*	*	*	*	*	-
Comparability																																																				
Study controls for age and sex	*	*	*	*	*	-	*	*	*	-	*	-	*	*	*	*	*	*	-	-	-	*	*	*	*	-	*	*	*	*	*	-	*	-	-	*	*	*	*	*	*	-	*	*	*	-	*	*	*	*	*
Study controls for others	*	*	*	*	*	-	*	*	*	*	*	*	-	*	*	*	*	*	-	*	-	*	*	*	*	-	*	*	*	*	*	-	*	-	-	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	-
Outcome																																																				
Statistical test†	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Assessment of outcome	*	*	✓	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*
Overall RoB Judgement[a]	G	G	G	G	P	P	G	P	G	P	G	G	G	G	G	F	F	F	P	G	P	P	G	G	G	P	G	G	G	G	G	P	F	P	P	G	G	F	G	G	P	G	G	P	G	G	G	P	G	G	G

Note. G = Good; F = Fair; P = Poor.

†Statistical test must be clearly described and appropriate before assessing the other items under the Outcome category. [a] Good rating is given when there are 3 to 4 stars in Selection category (Representativeness of Sample must be fulfilled) AND 1 to 2 stars in Comparability category AND 1 star in Outcome category; Fair rating is given when there are 2 stars in Selection category AND 1 to 2 stars in Comparability category AND 1 star in Outcome category; Poor rating is given when there are 0 to 1 star in Selection category OR 0 star in Comparability category OR 0 star in Outcome category.

"*" Study satisfies the criteria; "-" Study did not satisfy the criteria; "✓" Statistical test is clearly described and appropriate; "X" Statistical test is not described, incomplete or inappropriate